

Answer **all** the questions in the spaces provided.

For
Examiner's
Use

1 Determination of the concentration of sodium thiosulfate in solution Z

Sodium thiosulfate is commonly used in an alternative method for the extraction of gold from its ores, through the formation of a stable and soluble complex of gold(I) ions, $[\text{Au}(\text{S}_2\text{O}_3)_2]^{3-}$. One of your tasks is to determine the concentration of sodium thiosulfate in a commercially produced solution **Z** used in the extraction process.

Solution **Z** contains sodium thiosulfate in deionised water. You are not provided with solution **Z**, as it is too concentrated for use in this experiment. A diluted solution of **Z** has been prepared for you. This diluted solution is **FA 5**.

FA 1 is $0.0230 \text{ mol dm}^{-3}$ of aqueous potassium iodate(V), KIO_3

FA 2 is 0.50 mol dm^{-3} potassium iodide, KI

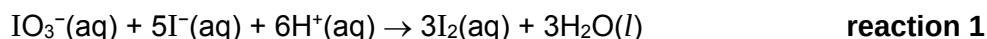
FA 3 is 0.75 mol dm^{-3} sulfuric acid, H_2SO_4

FA 4 is a solution of sodium thiosulfate, $\text{Na}_2\text{S}_2\text{O}_3$ of an unknown concentration

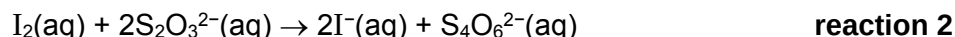
FA 5 a diluted solution of **Z**, in which 31.00 cm^3 of solution **Z** was made up to 250 cm^3 with deionised water in a volumetric flask

You are also provided with starch solution.

In acidic medium, iodate(V) ions oxidise iodide ions to iodine



If an excess of sodium thiosulfate is present in the reaction mixture, the iodine produced in **reaction 1** is immediately reduced back to iodide ions, as shown in **reaction 2**, and the dark brown colour of iodine does not form.



In the experiments that you will perform, insufficient volumes of the first sodium thiosulfate solution will be added initially, and some iodine will remain in the solution. The amount of iodine left-over can be determined by titration with the second solution of sodium thiosulfate, as shown in **reaction 2**.

You will prepare a number of reaction mixtures, each of which will contain:

- a fixed volume of **FA 1**,
- excess amounts of **FA 2** and **FA 3**,
- a different volume of **FA 4**.

All reaction mixtures will contain some iodine, produced in **reaction 1**, that will be titrated against the **FA 5** solution.

You are required to use a graphical method to determine the concentrations of sodium thiosulfate in **FA 4**, **FA 5** and solution **Z**.

(a) Titration of iodine produced against FA 5

Note: You will perform each titration only **once**. Great care must be taken to ensure that you do not overshoot the end-point.

You will be using two burettes in this task, and each of the burette will contain different sodium thiosulfate solutions, so care must be taken that the right one is used each time.

(i) Experiment 1

1. Fill the burette labelled "**FA 4**" with **FA 4** solution.
2. Fill the burette labelled "**FA 5**" with **FA 5** solution.
3. Using a pipette, transfer 25.0 cm³ of **FA 1** into a 250 cm³ conical flask.
4. From the burette labelled "**FA 4**", transfer 9.50 cm³ of **FA 4** to the same conical flask.

If you accidentally added an incorrect volume of **FA 4** for any of your titrations, record the actual volume of **FA 4** added in your table on the next page, and use that volume for the plotting of the graph later

5. Use appropriate measuring cylinders to add to this flask:
 - 10 cm³ of **FA 2**,
 - 10 cm³ of **FA 3**.
6. Titrate the iodine liberated with **FA 5** until the solution fades to a pale yellow colour.
7. Add about 1 cm³ of starch indicator to this flask. Continue adding **FA 5** until the blue-black colour **just** disappears.
8. Record your titration results in the section "Table of results" on page 3. Make sure that your recorded results show the precision of your working.

(ii) Experiment 2

9. Repeat steps 3 to 8 but add 22.00 cm³ of **FA 4** at step 4.

(iii) Experiment 3 to 5

10. Choose three further volumes of **FA 4** between 9.50 cm³ and 22.00 cm³.
11. Repeat steps 3 to 8 using each of your chosen volumes of **FA 4** at step 4.

Recording

Prepare a table in the space provided on the next page to record:

- the volumes of **FA 4** added for each experiment.
- the burette readings and the volumes of **FA 5** used for each experiment.

Table of results

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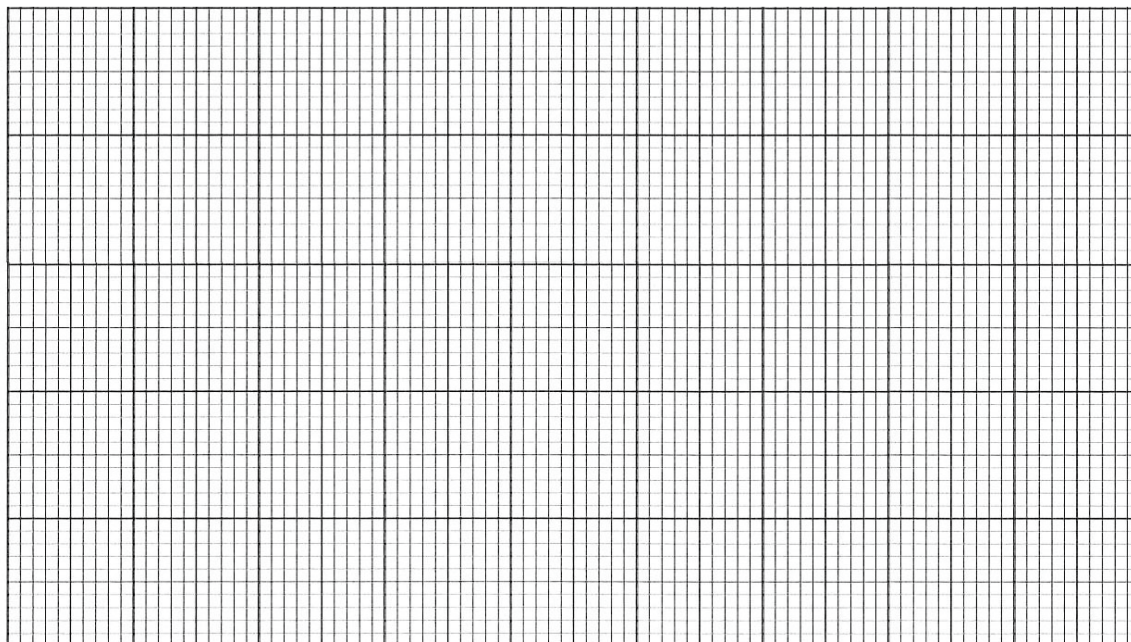
[4]

(b) Graphical determination of the concentration of sodium thiosulfate in FA 4, FA 5 and solution Z.

- (i) Plot a graph of the volume of **FA 5** added (*y-axis*) against the volume of **FA 4** added (*x-axis*) on the grid below. When choosing the scale, you should ensure that the graph line can be extrapolated to intersect both the *x-axis* and the *y-axis*.

Draw a straight line of best fit, taking into account all of your plotted points.

Extrapolate your graph line so that it intersects both the *x-axis* and the *y-axis*.



[3]

- (ii) It is possible to react all the iodine produced by the reaction between 25.0 cm³ of **FA 1** and 10 cm³ of **FA 2** and **FA 3** by using **FA 4** only.

From your graph, determine the volume of **FA 4** required to do this.

volume of **FA 4** required = [3]

- (iii) Calculate the gradient of your graph line, showing clearly how you did this.

gradient = [1]

(c) Calculations

- (i) The amount of iodate(V) ions, IO₃⁻, present in 25.0 cm³ of **FA 1** is 5.75 × 10⁻⁴ mol.

Use this value, together with data obtained from your graph in **(b)(i)**, to calculate the concentration of thiosulfate ions in **FA 4**.

[S₂O₃²⁻] in **FA 4** = [2]

- (ii) The concentration of S₂O₃²⁻ in **FA 4** and **FA 5** are related to the gradient of the graph obtained in **(b)(iii)** by the following equation:

$$[\text{S}_2\text{O}_3^{2-}]_{\text{FA 5}} = \frac{[\text{S}_2\text{O}_3^{2-}]_{\text{FA 4}}}{|\text{gradient}|}$$

Using the equation, or otherwise, calculate the concentration of thiosulfate ions in **FA 5**.

[S₂O₃²⁻] in **FA 5** = [1]

- (iii) Calculate the concentration of thiosulfate ions in the commercially produced solution **Z**.

[S₂O₃²⁻] in solution **Z** = [1]

- (d) A student suggested using burettes, rather than using measuring cylinders, to measure the volume of **FA 2** and **FA 3** used in (a) so as to improve on the accuracy of the titration data. State whether you agree with the student and briefly explain why.

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.....
.....
..... [1]

- (e) In a private institution, a laboratory assistant made the mistake in the preparation of the **FA 5** solution by diluting 21.00 cm³ of solution **Z** to 250 cm³, rather than the specified volume of 31.00 cm³.

State and explain how using this incorrect sample of **FA 5** will affect the gradient of your graph.

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.....
.....
..... [1]

[Total: 17]

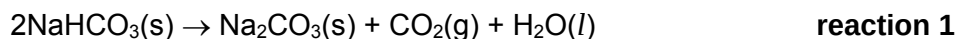
2 Determination of the enthalpy change for the thermal decomposition of sodium hydrogen carbonate

FA 6 is solid anhydrous sodium carbonate, Na_2CO_3

FA 7 is solid sodium hydrogen carbonate, NaHCO_3

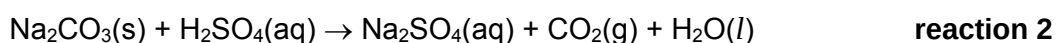
You will need access to the **FA 3** solution that you have used earlier.

Sodium hydrogen carbonate is commonly used as a raising agent in baking, as it undergoes thermal decomposition at temperatures above $80\text{ }^\circ\text{C}$, according to the equation:

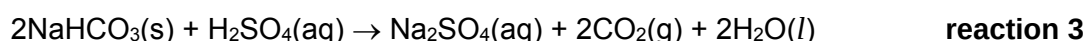


Due to the high temperature involved, the enthalpy change of this reaction cannot be measured directly using a coffee-cup calorimeter.

In the experiments that you will perform, the coffee-cup calorimeter will be used to determine the enthalpy change, ΔH_2 , for the reaction:



and the enthalpy change, ΔH_3 , for the reaction:



You will then use your results of your experiments to calculate the enthalpy change, ΔH_1 , for the thermal decomposition of sodium hydrogen carbonate.

(a) Determining enthalpy change for the reaction between Na_2CO_3 and H_2SO_4 , ΔH_2

1. Place the styrofoam cup inside a 250 cm^3 beaker to prevent it from toppling over.
2. Using a measuring cylinder, place 40 cm^3 of **FA 3** into the styrofoam cup.
3. Stir the solution in the cup gently with the thermometer, and measure its temperature. Record the initial temperature of **FA 3** in your table.
4. Weigh the weighing bottle labelled "**FA 6**" and its contents using a weighing scale, without the cap. The bottle has been pre-weighed to contain between 4.50 g and 5.00 g of solid sodium carbonate.
5. Pour **FA 6** from the weighing bottle into the styrofoam cup, place the lid onto the cup, and insert the thermometer through the lid.
6. Stir the mixture using the thermometer and record the maximum temperature reached.
7. Reweigh the weighing bottle.

(i) In an appropriate format in the space below, record:

- all mass readings, including the mass of **FA 6** added to the styrofoam cup
- all temperatures measured and the temperature rise.

[1]

- (ii) Calculate the heat change for **reaction 2**.

Assume that the specific heat capacity of the final solution is $3.75 \text{ J g}^{-1} \text{ K}^{-1}$ and that its density is 1.00 g cm^{-3} .

heat change = [1]

- (iii) By calculating the number of moles of each reagent that was added to the styrofoam cup, and with reference to the equation for the reaction, show which of the reagents, **FA 3** or **FA 6**, is added in excess.

[A_r : Na, 23.0; C, 12.0; O, 16.0]

reagent added in excess = [1]

- (iv) Calculate enthalpy change for **reaction 2**, ΔH_2 :



ΔH_2 = [1]

(b) Determining enthalpy change for the reaction between NaHCO_3 and H_2SO_4 , ΔH_3

1. Empty and rinse the styrofoam cup used in **(a)**, and place the cup back in the 250 cm³ beaker.
2. Using the same measuring cylinder used in **(a)**, place 40 cm³ of **FA 3** into the styrofoam cup.
3. Wash and dry the thermometer used in **(a)**, before using it to stir the solution in the cup gently, and measure its temperature. Record the initial temperature of **FA 3** in your table.
4. Weigh the weighing bottle labelled "**FA 7**" and its contents using a weighing scale, without the cap. The bottle has been pre-weighed to contain between 3.50 g and 4.00 g of solid sodium hydrogen carbonate.
5. Pour **FA 7** from the weighing bottle into the styrofoam cup, place the lid onto the cup, and insert the thermometer through the lid.
6. Stir the mixture using the thermometer and record the minimum temperature reached.
7. Reweigh the weighing bottle.

(i) In an appropriate format in the space below, record:

- all mass readings, including the mass of **FA 7** added to the styrofoam cup
- all temperatures measured and the temperature fall.

[3]

(ii) Calculate the heat change for **reaction 3**.

Assume that the specific heat capacity of the final solution is 3.75 J g⁻¹ K⁻¹ and that its density is 1.00 g cm⁻³.

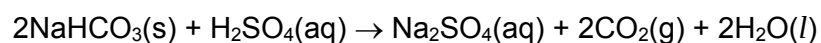
heat change = [1]

- (iii) By calculating the number of moles of each reagent that was added to the styrofoam cup, and with reference to the equation for the reaction, show which of the reagents, **FA 3** or **FA 7**, is added in excess.

[Ar: Na, 23.0; H, 1.0; C, 12.0; O, 16.0]

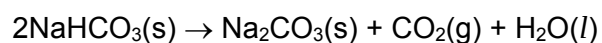
reagent added in excess = [1]

- (iv) Calculate enthalpy change for **reaction 3**, ΔH_3 :



$\Delta H_3 = \dots\dots\dots$ [1]

- (c) Use your answer to **(a)(iv)** and **(b)(iv)** to calculate enthalpy change for **reaction 1**, ΔH_1 :



$\Delta H_1 = \dots\dots\dots$ [4]

- (d) In (a), one of the significant sources of errors that will affect the accuracy of the enthalpy change for the reaction between Na_2CO_3 and H_2SO_4 , ΔH_2 , is that the heat capacity of the styrofoam cup is not taken into account.

Explain the effect of this error on your calculated value of ΔH_2 .

.....

.....

..... [1]

[Total: 15]

3 Planning

A sample of solid anhydrous sodium carbonate is believed to be contaminated with sodium chloride, NaCl .

The percentage purity of sodium carbonate in the sample can be determined by measuring the volume of carbon dioxide formed when sodium carbonate is reacted with an excess of sulfuric acid.



You are to plan an experiment that will enable you to:

- react a pre-weighed impure sample of sodium carbonate with a pre-determined volume of the **FA 3** solution that you have used earlier for Question 1 and 2,
- collect and measure the volume of the carbon dioxide gas evolved.

- (a) Draw a diagram of the apparatus that you would use to carry out this experiment.

[2]

- (b) By considering the capacity of the apparatus that you have drawn in (a) to measure the volume of the carbon dioxide gas, calculate the maximum mass of the impure sample of sodium carbonate that you would use in your experiment.

[A_r: Na, 23.0; C, 12.0; O, 16.0 and 1 mole of gas occupies a volume of 24.0 dm³ under the conditions present in the laboratory]

maximum mass of impure sample of sodium carbonate used = _____ [2]

- (c) A student suggested that 40.0 cm³ of **FA 3** solution is sufficient for the experiment. State whether you agree with the student, and explain your answer.

_____ [1]

- (d) State how you would ensure that all the carbon dioxide gas given off in the reaction between the impure sample of sodium carbonate and **FA 3** solution was collected and that none escaped from the apparatus.

_____ [1]

- (e) Other than this method that involves the collection of carbon dioxide gas, the percentage purity of sodium carbonate in the sample can also be determined using a titration.

In a typical experiment, a pre-weighed mass of the impure sample of sodium carbonate is dissolved in water and made up to 250 cm^3 in a volumetric flask, and 25.0 cm^3 of this solution is titrated against a standard solution of hydrochloric acid using methyl orange as indicator.

State which of the two methods you would expect to give the more accurate results. Explain your choice.

[2]

[Total: 8]

4 Inorganic Qualitative Analysis

You are provided with **FA 8**, which is a mixture of two solids, **FA 9**, which is soluble in water and **FA 10**, which is insoluble in water. Each contains **one cation** and **one anion** from the lists on page 17 and 18.

In this question you will perform a series of test-tube reactions, make observations and deduce the identities of the cations and anions present in **FA 9** and **FA 10**.

If the evolution of a gas is observed at any stage, the gas should be tested and identified.

You are advised that the reagent should be added gradually in all tests, with shaking after each addition.

(a) Tests on FA 8

Perform the test-tube experiments described below and record your observations in the spaces provided in the table.

	Test	Observations
(i)	Place all of the solid, FA 8 , into a 100 cm ³ beaker. Add 25 cm ³ of distilled water and warm to dissolve the FA 9 . Filter the mixture and use the filtrate for tests (a)(ii) to (a)(iv). Wash the residue, FA 10 , and use it for tests (a)(v) to (a)(vii).	

Tests on Filtrate, FA 9

	Test	Observations
(ii)	Place about 1 cm depth of FA 9 into a test-tube. Add a few drops of dilute sulfuric acid.	
(iii)	Place about 1 cm depth of FA 9 into a test-tube. Add a few drops of aqueous barium chloride, followed by a few drops of dilute hydrochloric acid.	
(iv)	Place about 1 cm depth of FA 9 into a test-tube. Add a few drops of aqueous silver nitrate. followed by a few drops of aqueous ammonia.	

Tests on Residue, FA 10

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	Test	Observations
(v)	Using a spatula, transfer as much of the residue on the filter paper as you can into a test-tube. Add dilute sulfuric acid drop-wise into the test-tube to dissolve the residue. Filter if necessary. Use the resultant solution for test (a)(vi) and (a)(vii).	
(vi)	Place about 1 cm depth of the filtrate from (a)(v) in a test-tube. Add aqueous sodium hydroxide drop-wise with shaking, until no further change is seen.	
(vii)	Place about 1 cm depth of the filtrate from (a)(v) into a test-tube. Add aqueous ammonia, drop-wise with shaking, until no further change is seen.	

[5]

- (b) Based on your observations to (a)(ii), suggest a possible identity of the cation in FA 9.

cation in FA 9: _____

[1]

- (c) From your remaining observations, state the identities of the one anion in both FA 9 and FA 10, and the one cation in FA 10.

In each case, give evidence to support your conclusion.

	identity		evidence
FA 9	anion:		
FA 10	cation:		
	anion:		

[3]

[Total: 9]

Qualitative Analysis Notes

[ppt. = precipitate]

For
Examiner's
Use**(a) Reactions of aqueous cations**

cation	reaction with	
	NaOH(aq)	NH ₃ (aq)
aluminium, Al ³⁺ (aq)	white ppt. soluble in excess	white ppt. insoluble in excess
ammonium, NH ₄ ⁺ (aq)	ammonia produced on heating	–
barium, Ba ²⁺ (aq)	no ppt. (if reagents are pure)	no ppt.
calcium, Ca ²⁺ (aq)	white ppt. with high [Ca ²⁺ (aq)]	no ppt.
chromium(III), Cr ³⁺ (aq)	grey-green ppt. soluble in excess giving dark green solution	grey-green ppt. insoluble in excess
copper(II), Cu ²⁺ (aq)	pale blue ppt. insoluble in excess	blue ppt. soluble in excess giving dark blue solution
iron(II), Fe ²⁺ (aq)	green ppt. insoluble in excess	green ppt. insoluble in excess
iron(III), Fe ³⁺ (aq)	red-brown ppt. insoluble in excess	red-brown ppt. insoluble in excess
magnesium, Mg ²⁺ (aq)	white ppt. insoluble in excess	white ppt. insoluble in excess
manganese(II), Mn ²⁺ (aq)	off-white ppt. insoluble in excess	off-white ppt. insoluble in excess
zinc, Zn ²⁺ (aq)	white ppt. soluble in excess	white ppt. soluble in excess

(b) Reactions of anions

<i>ions</i>	<i>reaction</i>
carbonate, CO_3^{2-}	CO_2 liberated by dilute acids
chloride, $\text{Cl}^-(\text{aq})$	gives white ppt. with $\text{Ag}^+(\text{aq})$ (soluble in $\text{NH}_3(\text{aq})$)
bromide, $\text{Br}^-(\text{aq})$	gives pale cream ppt. with $\text{Ag}^+(\text{aq})$ (partially soluble in $\text{NH}_3(\text{aq})$)
iodide, $\text{I}^-(\text{aq})$	gives yellow ppt. with $\text{Ag}^+(\text{aq})$ (insoluble in $\text{NH}_3(\text{aq})$)
nitrate, $\text{NO}_3^-(\text{aq})$	NH_3 liberated on heating with $\text{OH}^-(\text{aq})$ and Al foil
nitrite, $\text{NO}_2^-(\text{aq})$	NH_3 liberated on heating with $\text{OH}^-(\text{aq})$ and Al foil; NO liberated by dilute acids (colourless $\text{NO} \rightarrow$ (pale) brown NO_2 in air)
sulfate, $\text{SO}_4^{2-}(\text{aq})$	gives white ppt. with $\text{Ba}^{2+}(\text{aq})$ (insoluble in excess dilute strong acids)
sulfite, $\text{SO}_3^{2-}(\text{aq})$	SO_2 liberated with dilute acids; gives white ppt. with $\text{Ba}^{2+}(\text{aq})$ (soluble in dilute strong acids)

(c) Test for gases

<i>ions</i>	<i>reaction</i>
ammonia, NH_3	turns damp red litmus paper blue
carbon dioxide, CO_2	gives a white ppt. with limewater (ppt. dissolves with excess CO_2)
chlorine, Cl_2	bleaches damp litmus paper
hydrogen, H_2	"pops" with a lighted splint
oxygen, O_2	relights a glowing splint
sulfur dioxide, SO_2	turns aqueous acidified potassium manganate(VII) from purple to colourless

(d) Colour of halogens

<i>halogen</i>	<i>colour of element</i>	<i>colour in aqueous solution</i>	<i>colour in hexane</i>
chlorine, Cl_2	greenish yellow gas	pale yellow	pale yellow
bromine, Br_2	reddish brown gas / liquid	orange	orange-red
iodine, I_2	black solid / purple gas	brown	purple

END OF PAPER 4